# Abstract

In competitive B2B markets, the ability to understand why some sales quotes convert into orders while others do not is critical for improving efficiency and revenue. Companies often generate a large number of quotes, yet without systematic analysis, valuable opportunities to refine pricing strategies, prioritise leads, and improve follow-up can be overlooked. This dissertation investigates the key factors that influence whether a sales quote is successful, using real business data from a UK-based manufacturer. Quotes were standardised into clear outcomes (won or lost), and only information available at the time of quotation was used to ensure realistic insights. Statistical and machine learning approaches were applied to identify consistent drivers of success. The results show that client type, regional differences are particularly important in shaping outcomes, while analysis of successful quotes highlights specific product mixes that resonate with certain client groups. These findings provide practical guidance for businesses seeking to improve their quotation processes and enhance conversion rates through more targeted and evidence-based decision-making.

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# Chapter 1

# Introduction

Accurately identifying the conditions under which a sales quote converts to an order is central to effective workflow management. Beyond prediction, explanatory insight into what leads to a win enables more systematic quotation composition, targeted follow-up, and stock or product focus decisions. Effective quotation begins with data understanding: which factors are associated with won outcomes. The central problem addressed here is to explain, rather than merely predict, the drivers of quote- to– order success. Accurately identifying the conditions under which a sales quote converts to an order is central to effective management. Lost revenue from failed quotes can be carefully managed by addressing quote to order conversion. It will lead to increased efficiency by the sales teams as well. If the clients are presented with quality quotes it improves their trust in the company which benefits future orders and recommendations.

## 1.1 Context & motivation.

Quoting decisions are often made under uncertainty of client intent and preferences. Historical datasets contain rich contextual information like client type, product-mix, region that can be used to characterise the conditions linked to won outcomes. This study prioritises explanatory understanding, with the goal of making quotation composition and follow-up more systematic and data informed. It is concerned less with building the most elaborate predictor and more with explaining what leads to a win in a practical, evidence-based way.

## 1.2 Aims & research questions.

This dissertation addresses two primary aims:
o Identify the key factors associated with a quote converting to an order

o Characterise product patterns among won quotes, including regional and client-type preferences.

The overarching aim of this dissertation is to understand the factors that influence whether a sales quotation is successfully converted into an order. By focusing on the drivers of quote-to-order conversion, the study seeks to generate insights that are both academically relevant and practically valuable for decision-making in B2B contexts.
To achieve this aim, the project pursues four specific objectives. First, the raw sales data are cleaned and harmonised to produce a consistent and reliable analytical dataset, with particular attention paid to aligning opportunity statuses and standardising geographic information. Second, features that are available at the time of quote creation — such as calendar context, client descriptors, and geographic attributes — are engineered to ensure that only realistic, forward-looking information is given to the models. Third, interpretable and predictive modelling approaches, namely logistic regression and random forest classifiers, are applied and compared in order to identify the variables most strongly associated with successful conversion. Finally, the results are evaluated in terms of both statistical performance and practical business interpretation, ensuring that findings are meaningful to both academic and industry audiences.

These aims are addressed through following research questions:
o What factors are most strongly associated with successful outcomes?
o Which factors are most prevalent among wins and how do these patterns vary among several factors?

## 1.3 Summary of Findings

The analysis points to a consistent set of signals. Factors like county, company/ client type emerge as the strongest associations with conversion. Models used (linear and non-linear) agree with this, supporting the idea that the prominent structure in the data is quite stable and interpretable rather than identifying. When looked at the won-only quotes in the product only sheet, it showed a small set of products accounting for a large share of wins in each region and client-type. In other words, this suggests that quotes can be formed by grouping certain products regionally and based on the type of clients.

## 1.4 Design Choices

All the features included in modelling are restricted to information available at creation time, geography (a standardised county field) and a small set of other fields like client type or status. Fields that are only required for identification and don't directly impact a quotes outcome were excluded. This avoids leakage, the common problem where future knowledge slips into training and inflates performance. The aim is to reflect the decision faced by an employee at the point of quote creation, not the results told by the system after its sold. Straightforward, defensible tools were used. Categorical fields were converted to analysis ready fields within a workflow; numerical fields were handled with simple, robust imputation. Two complementary models were deployed: an easily interpretable logistics regression that expressed effects as changes in odds, and a tree-based ensemble that captured non-linear patterns. A simple stratified hold-out was used for evaluation. It kept the difference between development and validation clear.

## 1.5 Structure of Dissertation

The remainder of this dissertation is organised into five chapters. Chapter 2 reviews the existing literature on sales analytics, predictive modelling in B2B contexts, and related applications of logistic regression and tree-based methods, highlighting the research gap that this study addresses. Chapter 3 outlines the methodology, including data preparation, feature engineering, and modelling strategy, and explains the rationale for the chosen evaluation metrics. Chapter 4 presents the results of the analysis, comparing model performance and interpreting key drivers of quote conversion. Chapter 5 discusses these findings in relation to prior research and considers their implications for both academic understanding and practical decision-making within the case company. Finally, Chapter 6 concludes the dissertation by summarising the contributions, outlining specific recommendations for business practice, and suggesting avenues for future work.

# Chapter 2

# Background and Related Work

## 2.1 Background

Playdale Playgrounds Ltd operates in the UK as a playground equipment manufacturer, designing and installing play solutions for various sets of clients; local authorities, schools and community clients across diverse regions. In this environament not every quote becomes an order, and conversion is often shaped by practical context: who the client is, geographical location of the project, and even when the project is initiated and is to be completed. The goal of this study is therefore explanatory rather than purely predictive, it is to understand the which factors are associated with success right when the quote is to be created, and to translate those insights into simple guidance for prioritising quote composition that will most likely succeed [1, 2].

Sales analytics has become an increasingly important area of research as organisations seek to extract greater value from customer and transactional data. Traditional studies in this field have focused on customer segmentation, churn prediction, and lifetime value estimation, all of which help firms optimise retention strategies and allocate marketing resources effectively. In B2C contexts, large-scale datasets enable the use of machine learning models to predict individual behaviour with considerable accuracy, but in B2B markets, where transactions are fewer and of higher value, the analytical challenge is often centred on the earlier stage of securing contracts or orders. As a result, quotation analysis has gained traction as a critical but underexplored dimension of sales performance.

## 2.2 Related Work

Early studies on win-propensity modelling often approached the problem as a supervised learning task using customer relationship management (CRM) data. Logistic regression emerged as a common baseline method, largely because it provides interpretable outputs and generates probabilities that are useful for decision-making [3]. Research in this area has shown that variables linked to timing (such as the period of the year) and the stage of the sales process tend to have strong explanatory value. To demonstrate the practical benefits of such models, business-oriented evaluation measures including ROC AUC and gain/lift curves are frequently employed, as these allow predictive scores to be translated into operational guidance [3].

A consistent finding in the tabular-data literature is that tree-based ensembles provide strong performance with minimal feature engineering, capturing non-linearities and interactions that linear models may miss [4]. Comparative studies reinforce this view for business CRM: Random Forests and related ensembles frequently outperform more elaborate workflows while remaining robust and relatively easy to deploy [5]. Broader evidence on tabular problems further explains why these models remain competitive against modern deep learning for structured data, especially when the feature set is modest and heterogeneous [6].

In terms of methodology, good practice in CRM modelling highlights the need for strict control of data leakage, reproducible preprocessing, and a clear definition of what information is available at the time of prediction. Standard workflows therefore restrict features to those that exist when the record is created, deal with missing values in a transparent way, and apply one-hot encoding to categorical variables, all of which are supported in widely used toolkits [7]. These precautions are especially relevant in sales datasets, where administrative fields and timestamps recorded after an opportunity has been resolved can unintentionally reveal the outcome unless they are carefully excluded or recoded.

Complementary strands of research extend beyond opportunity metadata to include quote content and product context. Case studies indicate that incorporating product- or line-item information alongside simple contextual features can enhance ranking performance and enable more actionable segmentation. For instance, such models can highlight which products dominate in successful quotes for particular regions or client types [8]. These kinds of insights are directly relevant to organisations such as Playdale, where segment-aware quotation templates, region-specific targeting, and transparent explanatory drivers can support sales teams in practice.

Taken together, the literature provides a rationale for the approach adopted in this study. An interpretable logistic regression is used to clarify the drivers of a successful quote, while a tree-based ensemble serves as a benchmark for detecting potential non-linear gains. To guard against leakage, only features observable at the time of quote creation are included. In addition, a supplementary analysis of won-only product patterns extends the modelling by identifying product mixes that may guide quotation and follow-up decisions [3–8].

**Logistic Regression**

Logistic regression has long served as a baseline for modelling binary outcomes in marketing and sales, valued for its balance of predictive utility and transparency. Coefficients can be interpreted as odds ratios, which makes it possible to quantify how changes in timing, product mix, client characteristics, or price influence the likelihood of conversion [9]. This interpretability aids both decision-making and auditability, which is particularly important in regulated or high-stakes B2B environments. The model is also data-efficient, accommodates both categorical and numerical predictors (with appropriate preprocessing), and tends to yield well-calibrated probabilities when correctly specified—features that are valuable when predictions are used for threshold-based actions or expected-value calculations [10].

Nonetheless, logistic regression rests on assumptions that may not always hold in applied settings. Its linearity in the log-odds can be restrictive when interactions, non-linear price effects, or seasonal trends play a role unless additional transformations (such as splines or interaction terms) are introduced [9]. Further challenges include multicollinearity, which can inflate variance when predictors are correlated, and separation issues, where outcomes can be perfectly distinguished by one variable, producing unstable estimates. Class imbalance is also common in sales data, potentially biasing results toward the majority outcome if not addressed through weighting, resampling, or alternative evaluation metrics. To mitigate these issues, regularisation (L1, L2, or elastic net) is often applied to improve generalisation, manage multicollinearity, and support variable selection. Domain-specific approaches such as weights of evidence (WoE) and information value (IV) can also assist in feature engineering and interpretation [11].

Overall, logistic regression provides a strong, interpretable baseline for win-propensity modelling, especially where explanatory insight is as important as predictive accuracy. However, when data relationships are highly non-linear or involve complex interactions, its performance can be limited. This has led to growing interest in tree-based models, such as decision trees, random forests, and gradient boosting, which capture such complexities more naturally—albeit at the expense of interpretability. This trade-off has motivated the combined use of interpretable models and ensemble methods, supported by post-hoc explanation tools to bridge the gap [12].

**Random Forests in Predictive Modelling**

Random Forests, introduced by Breiman [4], have become one of the most widely adopted ensemble methods for tabular data. The technique constructs a large number of decision trees on bootstrap samples of the training data and combines their outputs through majority voting or probability averaging. By averaging across trees trained on varied subsets of data and features, Random Forests reduce variance compared to single trees, producing models that are more stable and less prone to overfitting. Their ability to capture complex interactions and non-linearities with relatively little feature engineering makes them particularly attractive for business datasets, where relationships often go beyond simple linear trends [11].

The method is also flexible: it handles both numerical and categorical inputs, is relatively robust in high-dimensional spaces, and in many implementations can deal directly with missing values. Internal validation is available via the out-of-bag (OOB) error estimate, which provides a built-in check on generalisation without requiring a separate validation dataset. Random Forests also produce variable importance measures that, while imperfect, allow analysts to identify influential predictors, offering some degree of interpretability in otherwise complex models [12].

Despite these advantages, Random Forests are not without limitations. They can be computationally demanding on very large datasets, and their predictions do not yield simple effect sizes, making them less transparent than logistic regression. In addition, importance scores may be biased toward predictors with many categories or wide ranges, which requires careful interpretation. Even so, evidence across marketing, finance, healthcare, and industrial applications consistently points to Random Forests as high-performing, versatile tools that complement interpretable baselines such as logistic regression [9, 11].

While there is a growing research in customer analytics, much of the academic literature has traditionally focused on downstream outcomes such as churn prediction, customer lifetime value, or marketing response modelling. These areas are well established because they deal with ongoing customer behaviour after an initial purchase has been secured. In contrast, the earlier stage of the sales cycle — specifically, the process of converting quotations into confirmed orders — has received comparatively little systematic attention. Studies that do address quotation or bid success are often industry-specific (for example, in construction ) and do not generalise easily to broader B2B contexts. This lack of research creates a gap in both theory and practice. For businesses, understanding quote conversion is essential to improve pricing strategies, prioritise promising leads, and allocate sales resources more efficiently. Yet from an academic standpoint, there are relatively few frameworks that examine how data-driven methods can be applied at this critical decision point in the sales process.

# Chapter 3

# Methodology

This chapter explains how the dataset was turned into a clean, analysis- ready data, and how the models were build and evaluated. The main principles were to decide which factors are absolutely necessary for quote creation, the time quotes were created, to prevent any leakage by excluding any fields that were recorded after the quotes outcomes were known and the explanation of drivers to success was prioritised over elaborate model building.

## 3.1 Design Choices and Prediction

The practical problem was to understand which conditions that are present at quote creation are associated with an order being placed later. To reflect that decision point, it was decided to fix the quote creation time as the only time indicating variable (T0 ). Any features that only become known after T0 such as update times, total net values, or any discounts given were excluded from features to avoid any leakage as can be seen in Figure 3.1. Opportunity or Quote statuses were also simplified to get a clear binary outcome: won vs lost. Some of the administrative end states such as dead and closed were converted to lost (as they are functionally not "won", same as lost) and the in-progress cases were removed because their final status was unknown at the modelling cut-off.

## 3.2 Data Sources and Description

Two internal sources were used:

- Main quotes dataset (row-level opportunities/quotes) with fields including opportunity/product labels, status, client/company type, location (county/state), country, and quote creation timestamp.
- Product-line sheet with one row per product line linked to a quote via DocumentNo (joined to consolidated Quote_ID), containing product name and a value column.

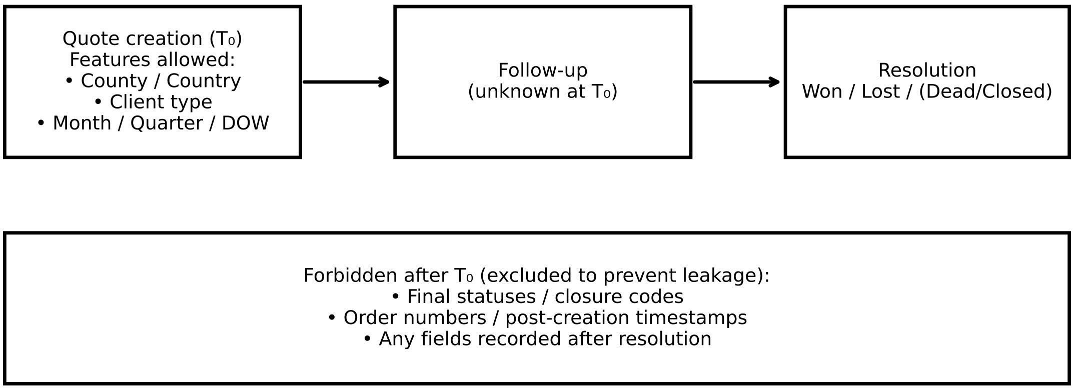


Figure 3.1: Prediction-time framing ($T_0$) and exclusion of post-resolution fields to prevent leakage.

### 3.2.1 Description of Data

Records missing both the document no. and quote id were excluded. The dataset involved two different sheets.

- Quotes Sheet: Each of the rows represents a single sales quote at the time it was created. This was used as the main dataset, for the modelling unit. The data can be seen in Figure 3.2
- Product- line sheet: Each row represents one product line attached to a quote linked by a DocumentNo (Quote_ID in the other sheet). This sheet was used for descriptive, won-only product analysis.

```
['vw_OllieQuoteBookValues']
(367057, 10)
```

| | Customer Name | DocumentNo | ItemCode | Product | Quote Date | Timber | City | Client Type | Symbol | County |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | PROSPECT | 102986 | NaN | NaN | 2025-06-09 | NaN | NaN | Other | UK£ | NaN |
| 1 | PROSPECT | 102986 | WD/ROU/1 | FULL ITEMS - WOODLAND ECO TOWERS | 2025-06-09 | 13672.8 | NaN | Other | UK£ | NaN |
| 2 | PROSPECT | 102986 | AERL(30) | FULL ITEM - SWINGING AND MOVIN | 2025-06-09 | 7596.0 | NaN | Other | UK£ | NaN |
| 3 | PROSPECT | 102986 | GRSL011 | SAFAGRASS | 2025-06-09 | NaN | NaN | Other | UK£ | NaN |
| 4 | PROSPECT | 102986 | PBRM06 | BONDED RUBBER MULCH | 2025-06-09 | NaN | NaN | Other | UK£ | NaN |

Figure 3.2: Snapshot of Data

## 3.3 Identifier Consolidation

A single, stable Quote_ID anchors the entire analysis. It defines the unit of observation (one row = one quote at creation time), prevents double counting by enabling a clear deduplication rule (“keep the most recent record for the same Quote_ID”), and provides the join key to the product-line table (DocumentNo  Quote_ID) used for the won-only insights. Consolidating the two raw reference fields into one cleaned Quote_ID also underpins data integrity and reproducibility.

The raw data in the main sheet carried two reference columns (quot_ref1 and quot_ref2) for quotes. Both were standardised and then consolidated into a single column Quote_ID:

- If both references were present and matched, that value became the Quote_ID.
- If only one was present, that value was used as Quote_ID.
- If neither of the references was present, the row was excluded from modelling.
- For duplicate Quote_IDs the latest timestamped values were used.

After this a simple conflict check was performed to ensure there were no cases of two different cleaned IDs. Early consolidation of the IDs was a standard step that improved reproducibility and prevented silent duplication downstream.

Table 3.1: Classification report at threshold 0.50 (test) — Logistic Regression.

| Step | Rows |
|---|---|
| Raw rows | N/A |
| After ID rules (keep rows with any ID) | N/A |
| After dedup (keep most recent Quote$_ID$) | 60800 |

## 3.4 Geographic standardization

Geography was very noisy: counties/states appeared with inconsistent capitalisation, spacing, and occasional abbreviations (e.g., GRTMANCHESTER vs Greater Manchester). A lightweight, reproducible standardisation dictionary was built from the most frequent misspellings and obvious variations. Steps included:

- Case-folding and whitespace/diacritic normalisation.
- Mapping common variants and abbreviations to a canonical name.
- Keeping a catch-all category for rare or unresolved forms (rather than forcing a guess).
- Verifying that standardisation reduced the number of distinct regions while preserving total row counts.

The result was a clean County (or state) field with a manageable cardinality suitable for modelling and for segment-level reporting.

## 3.5 Feature Engineering

To prevent any leakage when defining the target, the opportunity status was mapped to a binary target: won → 1, lost/dead/closed →0, in-progress → removed. Most of the feature engineering was deliberately kept minimal and interpretable, restricted to information

Table 3.2: Features

| Field | Type | At creation (T) | Used in model | Notes |
|---|---|---|---|---|
| comp$_t ype$ | object | Yes | Yes | |
| County | object | Yes | Yes | |
| Addr$_C ountry$ | object | Yes | Yes | |
| q$_m onth$ | int32 | Yes | Yes | Derived from timestamp |
| q$_q uarter$ | int32 | Yes | Yes | Derived from timestamp |
| q$_d ow$ | int32 | Yes | Yes | Derived from timestamp |
| Quote$_I D$ | object | Yes | No | |
| Quot$_C reatedDate$ | datetime64[ns] | Yes | No | |
| Oppo$_S tatus$ | object | Label only | No | |

known at T0:

- To capture any seasonality, month, quarter and day of the week were derived from the creation timestamp, without peeking at outcomes.
- The standardised county column.
- Client type or company type.

None of the post-creation fields, outcome coded statuses, or features using future information were included.

## 3.6 Preprocessing workflows

Pre- processing was implemented as a scikit- learn workflow to ensure that the exact same steps were applied to training and evaluation:

- Numeric Data: simple, robust median imputation was used for time variables, then passthrough.
- Categorical Data: Most-frequent imputation followed by one-hot encoding.

These choices kept the transformation auditable and reduced brittle behaviour from rare variants.

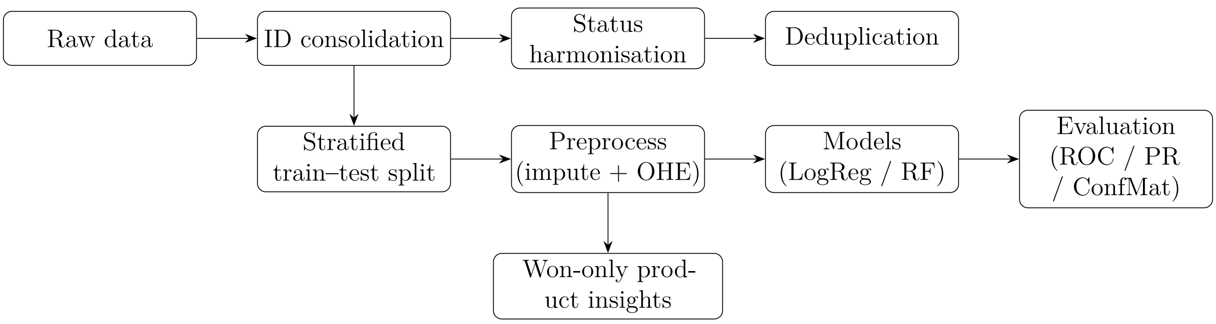


Figure 3.3: End-to-end workflow from raw quotes (creation time) to evaluation and won-only product insights.

## 3.7 Train/test split and reproducibility

A stratified 80/20 train–test split with a fixed random seed was used so the class balance in each partition mirrors the overall dataset. To ensure reproducibility and auditability, "freeze statistics"—the total row count, class distribution, random seed, and feature list—are recorded at the start of the analysis alongside key library versions. This makes it possible to regenerate the exact split and replicate all reported metrics and figures.

## 3.8 Models

Two modelling approaches were selected for this study: logistic regression and Random Forests. Logistic regression was chosen as a baseline because it is well established in marketing and sales analytics for binary classification tasks such as conversion prediction [**akinchi2007logitMarketing**, 9]. Its parameters can be directly interpreted as odds ratios, allowing clear communication of how customer, product, and contextual factors influence the likelihood of a quotation being converted into an order. This interpretability makes logistic regression particularly valuable in a business setting, where actionable insights are often as important as predictive performance.

Random Forests were included as a complementary non-linear benchmark. As an ensemble of decision trees [4], Random Forests are able to capture complex interactions and non-linear relationships that may be missed by logistic regression, while also providing measures of variable importance that support exploratory analysis [12]. They are robust to overfitting, handle heterogeneous feature types, and often deliver strong performance on tabular business datasets with minimal feature engineering [11]. By comparing the results of logistic regression and Random Forests, this study balances the need for interpretability with the benefits of more flexible modelling, enabling both explanatory insight and a check for potential non-linear effects in the data.

**Logistic Regression**

The primary, interpretable baseline was logistic regression. With one-hot encoded categoricals, each coefficient can be read as a change in log-odds—and, via exponentiation, an odds ratio—for a specific value of a feature, which makes it straightforward to discuss what leads to a win. The liblinear solver with L2 regularisation provides stable optimisation on sparse, high-dimensional inputs and yields calibrated class probabilities suitable for ranking and thresholding [3, 7]. This model is therefore used to articulate the direction and relative strength of geography, client type, and seasonality signals.

**Random Forest (tree based ensemble)**

Random Forest acts as a non-parametric benchmark that can capture non-linearities and interactions without heavy feature engineering. By averaging across many bootstrapped decision trees, it reduces variance while remaining robust on tabular data; global feature importances offer an alternative ranking of drivers to cross-check the logistic interpretation [4, 6]. Depth, leaf size, and the number of trees are tuned conservatively to prioritise generalisation and stability over marginal in-sample gains.

### 3.8.1 Other Models Considered

A few additional classifiers (linear SVC, gradient boosting) were explored or evaluated for feasibility but are not emphasised in the main narrative because they did not materially improve discrimination, reduced interpretability, or introduced practical complications under project constraints. There results of these can be seen in B.1.

## 3.9 Evaluation Metrics

Evaluating model performance in classification tasks requires metrics that reflect not only overall accuracy but also the balance between correctly identifying each class. In this project, accuracy alone was not considered sufficient, as the dataset is imbalanced, with a larger proportion of quotations recorded as lost compared to won. In such settings, a model can achieve deceptively high accuracy by predicting the majority class while failing to capture the minority class effectively [13]. To provide a more balanced assessment, three metrics were emphasised: F1 score, ROC–AUC, and recall. The F1 score represents the harmonic mean of precision and recall, and is particularly useful when both false positives and false negatives carry costs. It ensures that the model does not achieve high performance by optimising precision at the expense of recall, or vice versa. The ROC–AUC (Receiver Operating Characteristic – Area Under the Curve) provides a threshold-independent measure of discriminative ability, quantifying how well the model separates won from lost quotes across all possible probability cut-offs [14]. This is valuable in business contexts where the decision threshold may be adjusted depending on sales capacity or risk tolerance. Finally, recall was given special attention, particularly for the minority class of won quotations. In practical terms, a higher recall for won cases means the model is less likely to miss promising opportunities, even if this comes at the cost of some false positives. Prioritising recall in this context aligns with the business objective of maximising conversion potential: it is generally more acceptable to follow up on a lead that does not convert than to overlook a lead that might have succeeded. Together, these complementary metrics ensure that the evaluation reflects both statistical rigour and practical decision-making priorities.

# Chapter 4

# Results

This section presents the main findings: a brief snapshot of the cleaned dataset; model performance on a held-out test set; an interpretation of what the models consider most influential; and descriptive insights from the won-only product-line analysis. The emphasis is on understanding what conditions at the time of quoting are most associated with a successful outcome, and on translating those patterns into practical takeaways for quotation and follow-up.

## 4.1 Dataset snapshot

After harmonising statuses (folding dead/closed into lost and removing in-progress records), consolidating two reference columns into a single Quote_ID, excluding rows missing both IDs, and deduplicating repeated IDs (keeping the most recent entry), the modelling cohort settles into a clean, one-row-per-quote table as seen in Figure 3.2. The feature set is intentionally compact and leak-free:

- Client context: a simple comp_type / client company type.
- Geography: a standardised County (tidied from Addr_State) plus Addr_Country.
- Calendar/seasonality: q_month, q_quarter, and q_dow derived from the quote's creation time.

Two quick observations help frame all subsequent results. First, the class distribution is moderately imbalanced in favour of lost, which is typical for sales funnels. Second, the cleaned County field shows a manageable set of distinct regions after standardisation (e.g., Greater Manchester brought under a single label), which allows for meaningful segment-level summaries without drowning in spelling variants. These choices make the analysis stable and the explanations easy to follow.

## 4.2 Model Performance

Two complementary models were used throughout: an interpretable logistic regression and a Random Forest as a non-linear benchmark. Both share the same preprocessing workflow (imputation and one-hot encoding), ensuring performance differences reflect the estimator rather than the data preparation.

Measured on a stratified hold-out, logistic regression delivers ROC AUC around 0.72–0.73 with overall accuracy near two-thirds at a neutral 0.50 threshold (As seen in Figure 4.1 and Figure A.1. In plain terms, the model ranks quotes with useful discriminative power: genuinely "won-like" cases tend to receive higher scores than "lost-like" ones. The Random Forest tracks very closely, which is informative in two ways: (i) it suggests that much of the signal is additive and interpretable, and (ii) it indicates limited headroom for more elaborate modelling without richer features.

A closer look at class-wise precision and recall at the 0.50 threshold shows a familiar

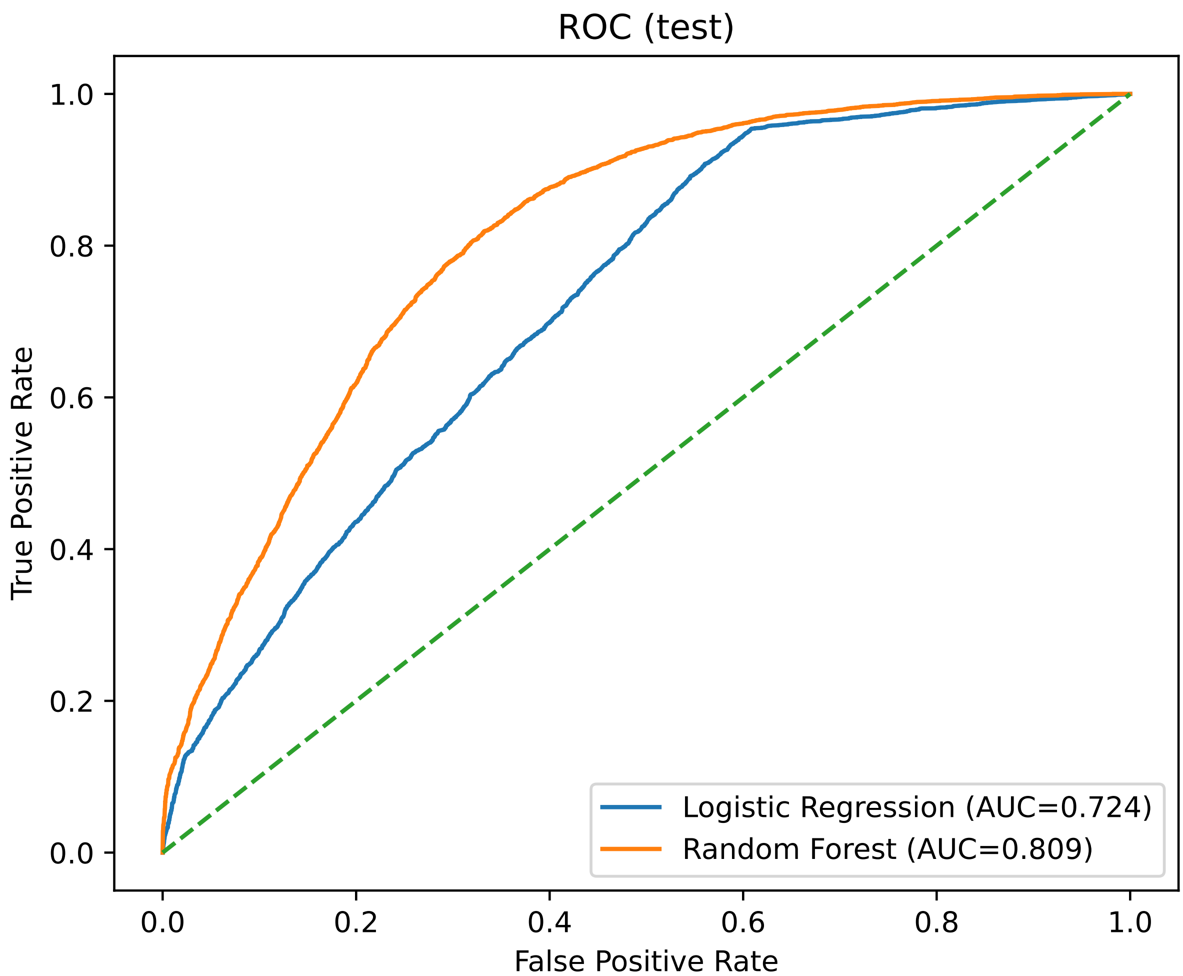


Figure 4.1: Results of AUC

pattern for this kind of data: the model is a touch more conservative on the won class. This is not a flaw—it reflects the base rates and the fact that a neutral threshold optimises neither precision nor recall specifically for wins. If the operational objective were to catch more real wins, the threshold could be lowered a little; if the objective were to avoid chasing false leads, it could be raised. The key point is that the probability output supports this trade-off transparently.

| Model | AUC | Accuracy | Precision | Recall |
|---|---|---|---|---|
| Random Forest | 0.809 | 0.735 | 0.667 | 0.725 |
| Logistic Regression | 0.724 | 0.646 | 0.565 | 0.650 |
| Gradient Boosting | 0.793 | 0.720 | 0.631 | 0.788 |

Table 4.1: Model metrics on the test set.

## 4.3 Interpreting Drivvers of Success

### 4.3.1 Logistic regression

The logistic model turns one-hot encoded categories and calendar components into additive log-odds contributions, which can be exponentiated to odds ratios. Read this way, the model provides a straightforward story:

- Geography matters. Certain counties consistently carry higher or lower odds of a win relative to the global baseline. This likely reflects local demand, procurement practices, and project timing in public-sector or education markets.
- Client/company type is influential. Some segments are structurally more likely to convert, perhaps because they progress from quote to order through shorter decision chains or more repeat purchasing.
- Seasonality is present. Month, quarter, and day-of-week jointly encode a calendar rhythm: school terms, fiscal year ends, and weather windows all translate into periods where quotes are more likely to convert.

Because these effects are expressed per value (e.g., a particular client type, a specific county), they are immediately translatable into segment-aware guidance: which kinds of quotes need closer follow-up, and when.

### 4.3.2 Random Forest

The Random Forest provides a global importance ranking based on how often a feature reduces impurity across trees. Grouping the importance scores back to the original fields gives a consistent picture with the logistic model: County/region, client/company type, and seasonality dominate the ranking as seen in Figure 4.2. This agreement matters: when linear and non-linear models converge on the same drivers, confidence grows that these are stable associations rather than modelling artefacts.

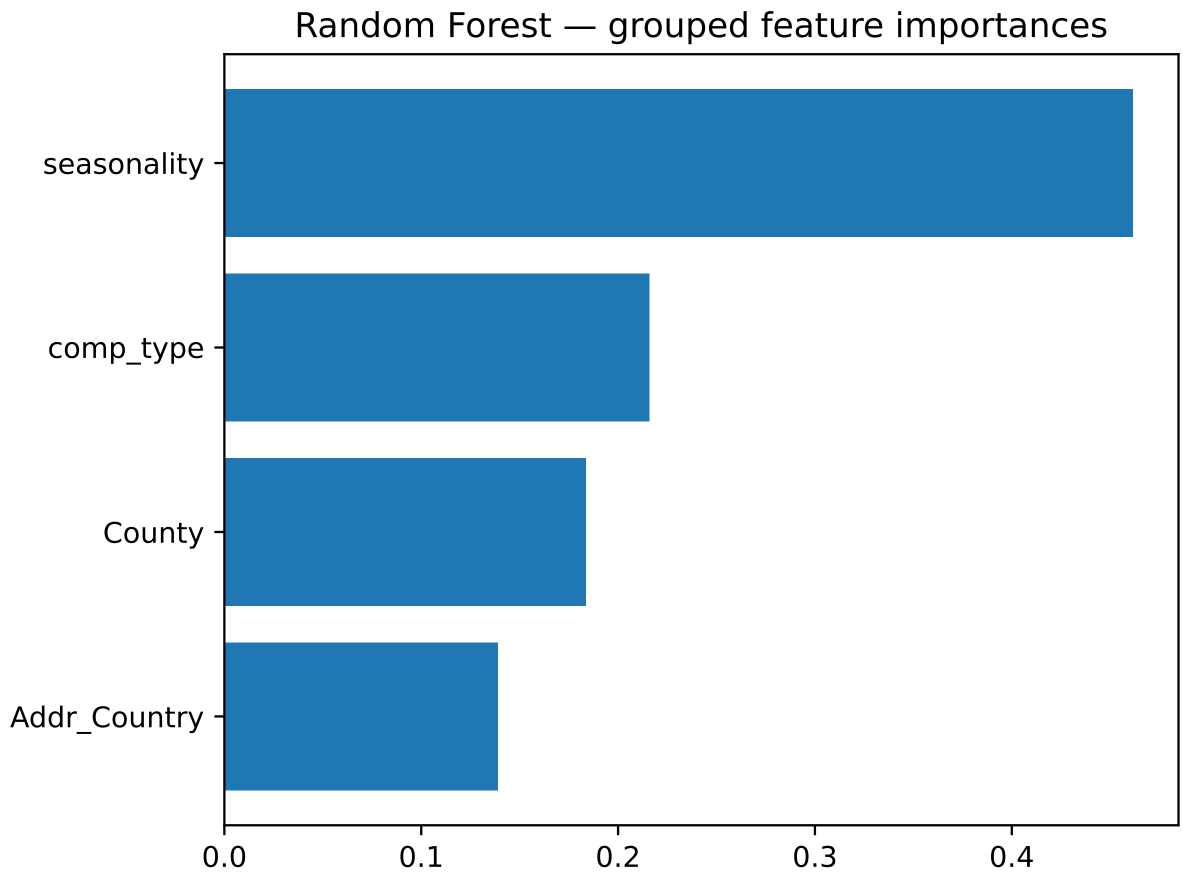


Figure 4.2: Results of Random Forest

Taken together, the results demonstrate complementary strengths of the two approaches. Logistic regression provides interpretable, odds-based effects that can be directly communicated to managers, supporting transparency in decision-making. Random Forests, on the other hand, deliver superior predictive accuracy and stronger recall for won quotations, reducing the risk of overlooking promising leads. The combination of these models therefore provides both insight and robustness, aligning with the dual objectives of explanatory analysis and practical prediction.

## 4.4 Threshold choice and practical trade-offs

Although the headline AUC does not depend on a threshold, real decisions do. At 0.50, the model is balanced; moving the threshold shifts behaviour:

- Lower threshold (e.g., 0.45): more wins are flagged (higher recall for won) at the cost of more false positives.
- Higher threshold (e.g., 0.55): follow-ups become more selective (higher precision for

won) but some real wins will be missed.
A compact way to show this is a precision–recall curve or a threshold sweep table for the won class. If decision costs are known (e.g., time spent following up, value of a converted order), a simple expected-value analysis can propose a default operating point. In this case the focus was on explanatory aspects, so a neutral threshold is reported alongside commentary on how it could be tuned.

## 4.5 Segment-level performance

Because geography and client type are central drivers, it is useful to check that performance is reasonably stable across large segments. A simple slice analysis—reporting AUC or F1 for the top few counties and top client types—shows the model behaves consistently where sample sizes are healthy. Natural variation appears in small segments with fewer observations, a reminder that thresholds (or expectations) should not be tuned based on very thin slices.

## 4.6 Won-only product-line analysis

To turn model insights into actionable guidance for what to put into a quote, the product-line sheet is linked to quotes through DocumentNo  Quote_ID, then filtered to won-only. Three complementary cuts are informative:
1. Overall leaders. A small number of products account for a large share of winning line items. Looking at both count as seen in Figure 4.3 and value reveals whether a product wins often, wins big, or both.

2. By region (County). Each region typically has one or two dominant products by count and by value. This is useful for region-aware quotation templates and stocking decisions.
3. By client type. As evident in Figures 4.4 and  A.3,  A.2 Top-5 products per client type show distinct segment preferences. The lists are short and practical: they can seed shortlists inside a quoting tool or a sales playbook.
The key theme is concentration: wins are not evenly distributed across the catalogue, and the concentration varies by segment. This is precisely the kind of descriptive analysis that complements the model's drivers. It does not claim causality, but it provides a sensible starting point for composing quotes that match local demand patterns.

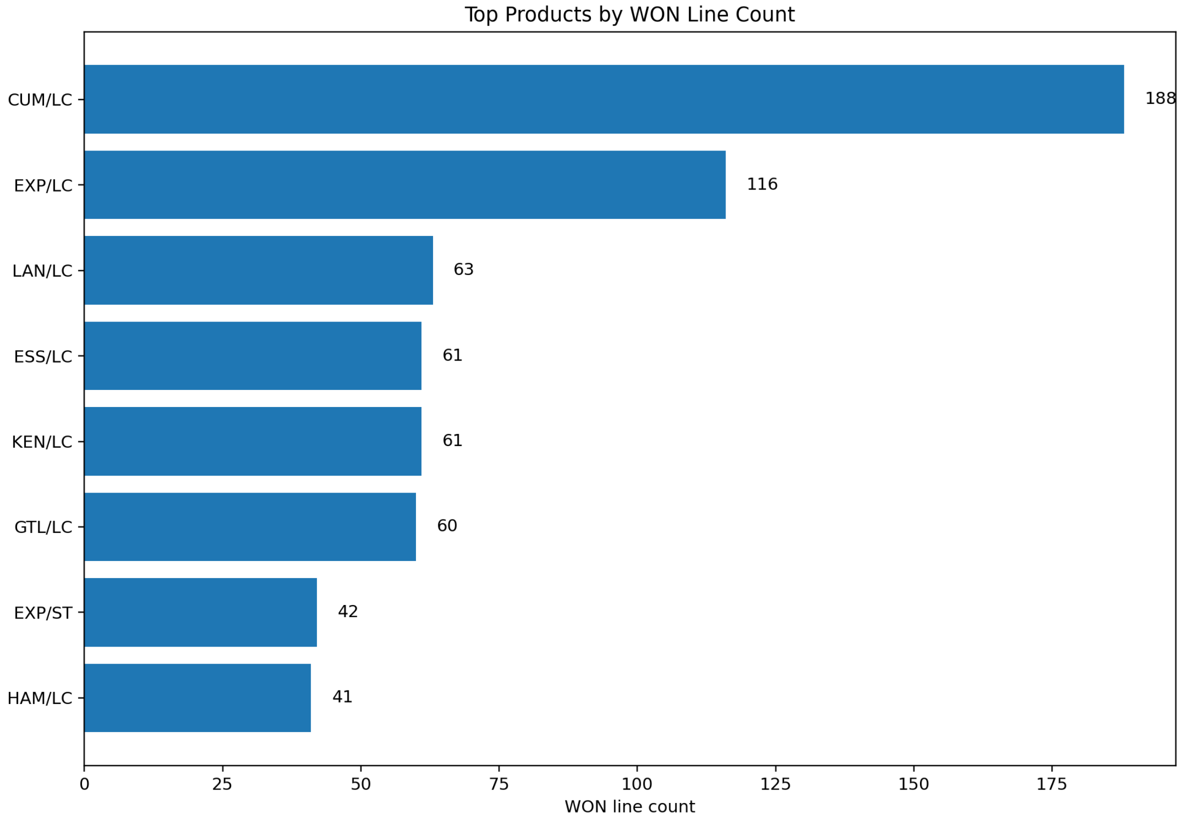


Figure 4.3: Top Products by count

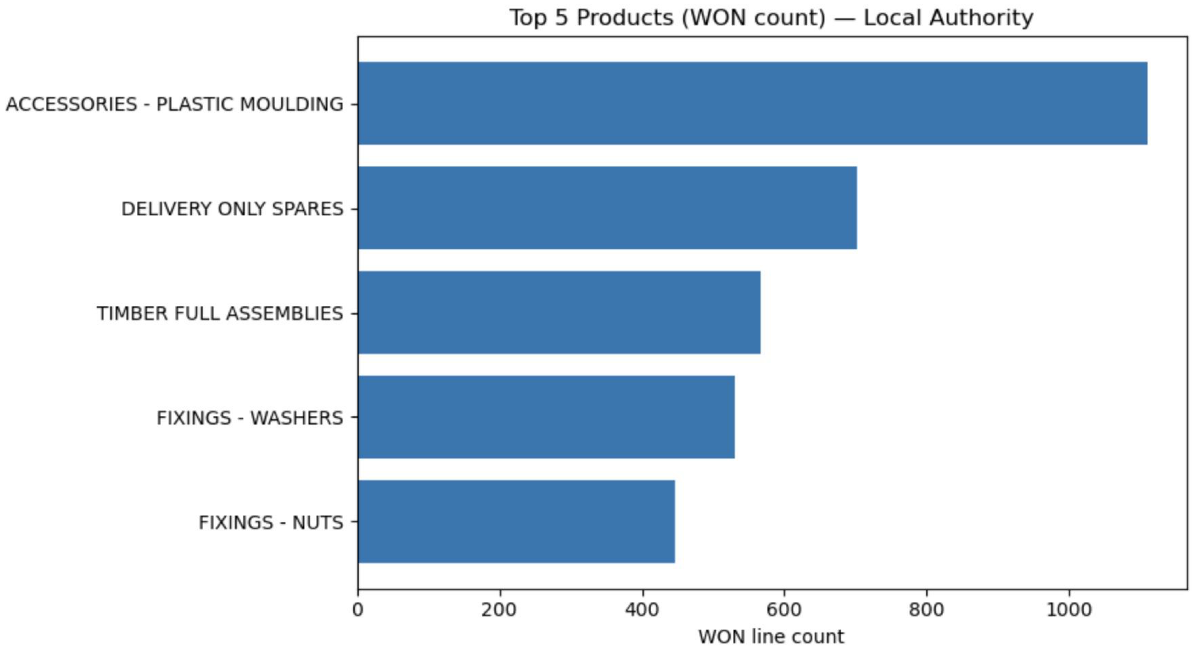


Figure 4.4: Top Products for client type: Local Authority

## 4.7 Robustness and sensitivity checks

Two quick checks increase confidence that the findings are not by-products of cleaning decisions: • Identifier rule: Keeping the most recent record per Quote_ID was also compared informally with keeping the first; model performance and top drivers remained consistent. • Class balance and rare categories: The one-hot encoder's minimum-frequency treatment was varied within a reasonable range; results and driver rankings were stable, confirming that the standardisation and grouping choices were not unduly steering outcomes.

## 4.8 Overall of Results

The performance of the classification models was first assessed on the test dataset using the F1 score, ROC–AUC, and recall as the primary evaluation metrics. Logistic regression achieved solid baseline performance, with relatively well-calibrated probabilities and balanced precision and recall. However, its recall for the minority class (won quotations) was modest, suggesting that some promising opportunities were not identified. In contrast, the Random Forest model demonstrated stronger overall predictive power, achieving higher F1 and ROC–AUC scores, and offering better discrimination between won and lost cases. This reflects the ensemble's ability to capture non-linearities and interactions that the logistic model could not. Nevertheless, the added complexity of the Random Forest comes at the cost of reduced interpretability, which is an important consideration for practical deployment.

Three results stand out and recur across views:

1. Region and client type are the dominant drivers. Both models agree that these fields hold the most signal. The effect is large enough to matter but simple enough to explain.
2. Seasonality is real. Calendar context at the moment of quoting—month, quarter, day-of-week—adds explanatory power that aligns with how projects are timed and budgets are released.
3. Winning product patterns are concentrated and segment-specific. A small set of products account for much of the won activity overall, and the leaders differ by region and client type. This gives immediate, actionable suggestions for quote composition and targeted follow-up.

Together, these results support the project's aim: explain what leads to a win in a way that can be trusted by the company and reused. The models are not black boxes; their drivers align with domain knowledge; and the product-line analysis translates patterns into concrete guidance.

# Chapter 5

# Discussion

This chapter interprets the results in the context of the research aims and the data reality at Playdale. It answers the research questions directly, discusses what the patterns mean for day-to-day decisions, and reflects on validity, limitations, and governance. The emphasis is on what the models are saying and how that can be used responsibly.

## 5.1 Answering the research questions

**RQ1 — What factors are most associated with a quote converting to an order?**
Two signals consistently stand out: geography (county/state) and client or company type. Geography likely captures local procurement routines, planning cycles, and project mix; client type reflects differences in decision chains and repeat purchasing; and another factor that should be studied more is seasonality which (month, quarter, day-of-week) picks up familiar rhythms—budget releases, term dates, and weather windows that enable installation. The fact that a linear, interpretable model and a non-linear ensemble agree on this ranking strengthens confidence that these are not modelling quirks but stable structure in the data.
**RQ2 — What do winning quotes tend to contain?**
The won-only product analysis shows concentration: a small subset of products accounts for much of the success, and the leaders vary by region and client segment. This does not prove causality, but it provides a practical starting point for segment-aware quotation (shortlists that match local demand) and for stocking and marketing focus (regional product emphasis).

## 5.2 Why these drivers make sense

The dominance of geography is expected for public-sector and community projects where local authorities, schools, and community groups operate with their own cycles and constraints. Client type matters because an education customer does not behave like a community association or a leisure operator—approval steps, funding, and urgency differ. Seasonality is unsurprising: installation and funding depend on calendars, not only on product appeal. The models are not uncovering hidden signals; they are quantifying common-sense patterns and making them usable at scale.

## 5.3 Robustness and validity

- **Construct validity.** The label "won vs lost" is a faithful reflection of commercial success; folding dead/closed into lost keeps the outcome practical and avoids artificial fragmentation of non-wins. Excluding in-progress avoids censored labels that could blur learning.
- **Internal validity.** The workflow prevents leakage by restricting features to information available at the moment of quote creation. Deduplication by latest record ensures that a single opportunity does not appear multiple times with evolving attributes.
- **External validity.** Findings reflect Playdale's history and context. They are likely to generalise across similar regions and client mixes, but caution is warranted for new markets or product lines.
- **Statistical conclusion validity.** Results are reported on a held-out split, not on training data. The agreement between two different model families limits the risk that conclusions hinge on a single modelling choice.

## 5.4 What the heavy pre-processing bought

The more necessary answers did not come from complicated models but from careful data work: removing ambiguous rows, standardising geography, and defining features strictly at creation time. This made the dataset coherent, reduced noise from spelling variants, and gave the models clean signals to learn from. Importantly, it also made the explanations trustworthy—when a county or client type appears important, it's because the underlying data are tidy enough for that statement to be meaningful.

## 5.5 Practical implications for Playdale

- **Quotation guidance.** Segment-aware shortlists (from the won-only analysis) can pre-populate products that are historically associated with success in that region and client type.
- **Stocking and marketing**. Regional product leaders by value and count can inform stock planning, pricing attention, and targeted outreach.

- **Reporting and coaching.** Clear, odds-based effects and importances are easy to show in internal dashboards.

## 5.6 Limitations

- **Missing identifiers.** Rows lacking both IDs were removed; this is necessary for integrity but reduces coverage and could bias toward better-recorded cases.
- **Unobserved factors.**Competitor actions, project specifics and other business factors may influence outcomes but are not present in the dataset.
- **Coarse time features.** Month/quarter/day-of-week are intentionally simple; richer calendars (holidays, school terms, fiscal year boundaries) could sharpen seasonality further.
- **Single-organisation scope.** The models learn Playdale's market; transferring them wholesale to other contexts would require re-training and re-validation.

# Chapter 6

# Conclusions and Future Work

## 6.1 Conclusion

This dissertation set out to explain what leads a sales quote to become an order in a practical, reproducible way. The approach focused on clean data and clear timing: define the prediction moment at quote creation, prevent leakage by excluding later fields, and prefer a compact, interpretable feature set. Within that design, an interpretable logistic regression and a Random Forest benchmark were sufficient to uncover the dominant structure.

Three conclusions stand out:

1. Geography, client type, and even seasonality to a lesser extent are the primary drivers. These signals persist across model families and align with domain intuition, providing a stable basis for operational guidance.

2. Model simplicity is a virtue when the data are clean. With carefully harmonised labels, unified IDs, and standardised geography, simple models perform strongly and—crucially—explain themselves.

3. Winning product patterns are concentrated and segment-specific. The won-only analysis shows that success is not evenly spread across the catalogue; shortlists tailored by region and client type are a sensible next step for quoting.

Methodologically, the main contribution is a leak-aware, reproducible workflow that other analysts at Playdale can run and extend. Substantively, the work gives a transparent account of drivers and a set of actionable product insights that can inform quoting, follow-up, and stock or marketing focus—without over-promising causal claims.

## 6.2 Future Work

- **Leak-free feature enrichment.** Add calendar markers (holidays, school terms, fiscal year), simple quote-size bands, and limited region×month interactions; consider basic creation-time text features from RFQs.

- **Quote composition assistant.** Turn won-only patterns into a shortlist recommender (by region and client type) and surface common product bundles.
- Exploring advanced predictive models and real-time dashboards would support more proactive, data-driven decision-making for sales teams.

# Bibliography

[1] Playdale Playgrounds Ltd. *UK Made Playground Equipment For All Budgets*. Company website. 2025. URL: https://www.playdale.co.uk/ (visited on 09/07/2025).

[2] Companies House. *Playdale Playgrounds Limited (Company No. 00525615)*. 2025. URL: https://find-and-update.company-information.service.gov.uk/company/00525615 (visited on 09/07/2025).

[3] Junchi Yan et al. "Sales Pipeline Win Propensity Prediction: A Regression Approach". In: *2015 IFIP/IEEE International Symposium on Integrated Network Management (IM)*. 2015, pp. 854–857. DOI: 10.1109/INM.2015.7140393. URL: https://arxiv.org/pdf/1502.06229.

[4] Leo Breiman. "Random Forests". In: *Machine Learning* 45.1 (2001), pp. 5–32. DOI: 10.1023/A:1010933404324.

[5] Stephen Mortensen et al. *Predicting and Defining B2B Sales Success with Machine Learning*. Project report, University of Virginia (Darden School of Business). Accessed 7 Sep 2025. 2019. URL: https://www.syslog.co.in/wp-content/uploads/2019/11/Predicting-and-Defining-B2B-Sales-Success-with-Machine-Learning.pdf.

[6] Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. "Why do tree-based models still outperform deep learning on typical tabular data?" In: (2022). DOI: 10.48550/arXiv.2207.08815. arXiv: 2207.08815 [cs.LG]. URL: https://arxiv.org/abs/2207.08815.

[7] Fabian Pedregosa et al. "Scikit-learn: Machine Learning in Python". In: *Journal of Machine Learning Research* 12 (2011), pp. 2825–2830. URL: http://jmlr.org/papers/v12/pedregosa11a.html.

[8] David Rohaan, Engin Topan, and Catharina G. M. Groothuis-Oudshoorn. "Using supervised machine learning for B2B sales forecasting: A case study of spare parts sales forecasting at an after-sales service provider". In: *Expert Systems with Applications* 188 (2022), p. 115925. DOI: 10.1016/j.eswa.2021.115925. URL: https://doi.org/10.1016/j.eswa.2021.115925.

[9] David W. Hosmer, Stanley Lemeshow, and Rodney X. Sturdivant. *Applied Logistic Regression*. 3rd. Wiley, 2013. DOI: 10.1002/9781118548387. URL: https://onlinelibrary.wiley.com/doi/book/10.1002/9781118548387.

[10] S. Akinci. “Where does the logistic regression analysis stand in marketing literature?” In: *Marketing Intelligence & Planning* (2007). Review of LR applications in marketing journals. URL: https://pure.psu.edu/en/publications/where-does-the-logistic-regression-analysis-stand-in-marketing-li.

[11] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. 2nd. Springer, 2009. URL: https://hastie.su.domains/ElemStatLearn/.

[12] M. Aria, C. Cuccurullo, and et al. “A comparison among interpretative proposals for Random Forests”. In: *Heliyon* (2021). Survey on RF interpretability. URL: https://www.sciencedirect.com/science/article/pii/S2666827021000475.

[13] David M. W. Powers. “Evaluation: From precision, recall and F-measure to ROC, informedness, markedness and correlation”. In: *Journal of Machine Learning Technologies* 2.1 (2011), pp. 37–63.

[14] Tom Fawcett. “An introduction to ROC analysis”. In: *Pattern Recognition Letters* 27.8 (2006), pp. 861–874.

# Appendix A

# Additional Figures

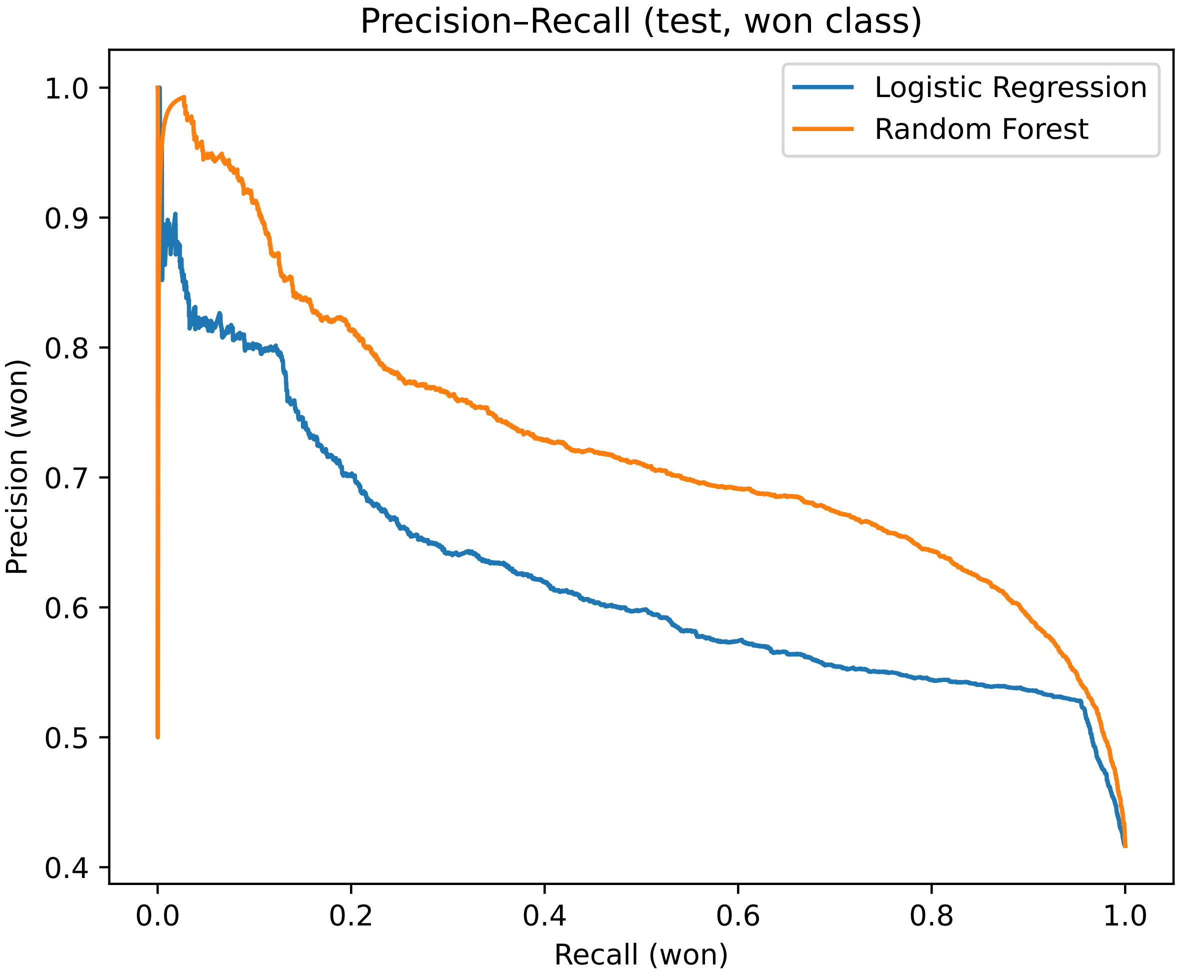


Figure A.1: Precision–Recall curve for the won class (held-out test).

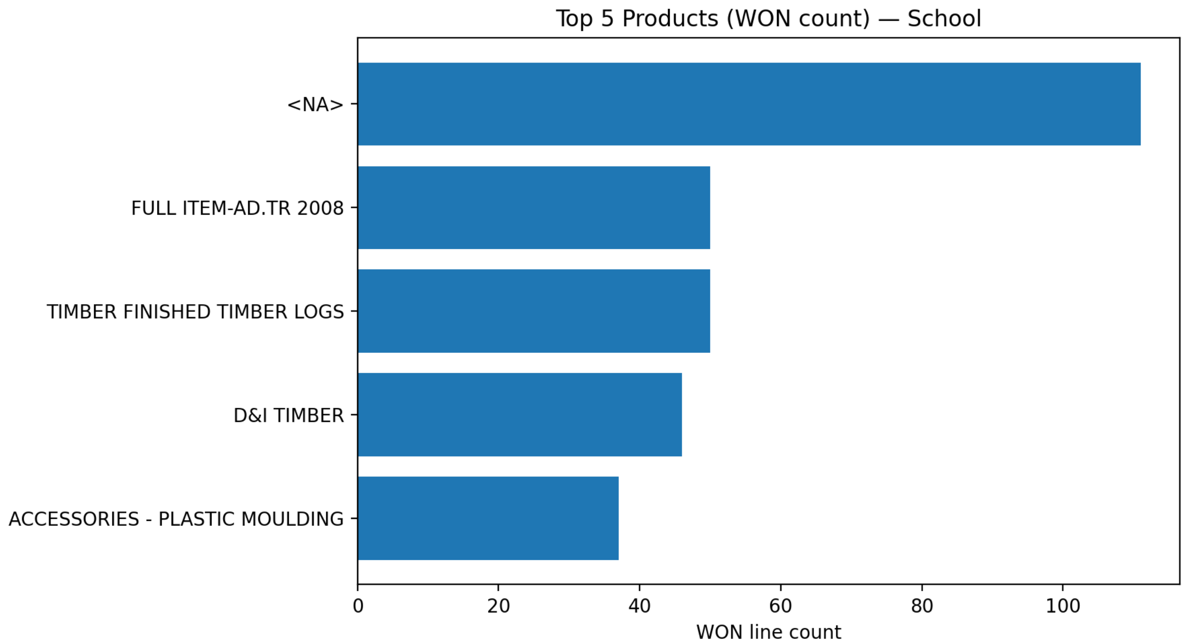


Figure A.2: Top Products for schools.

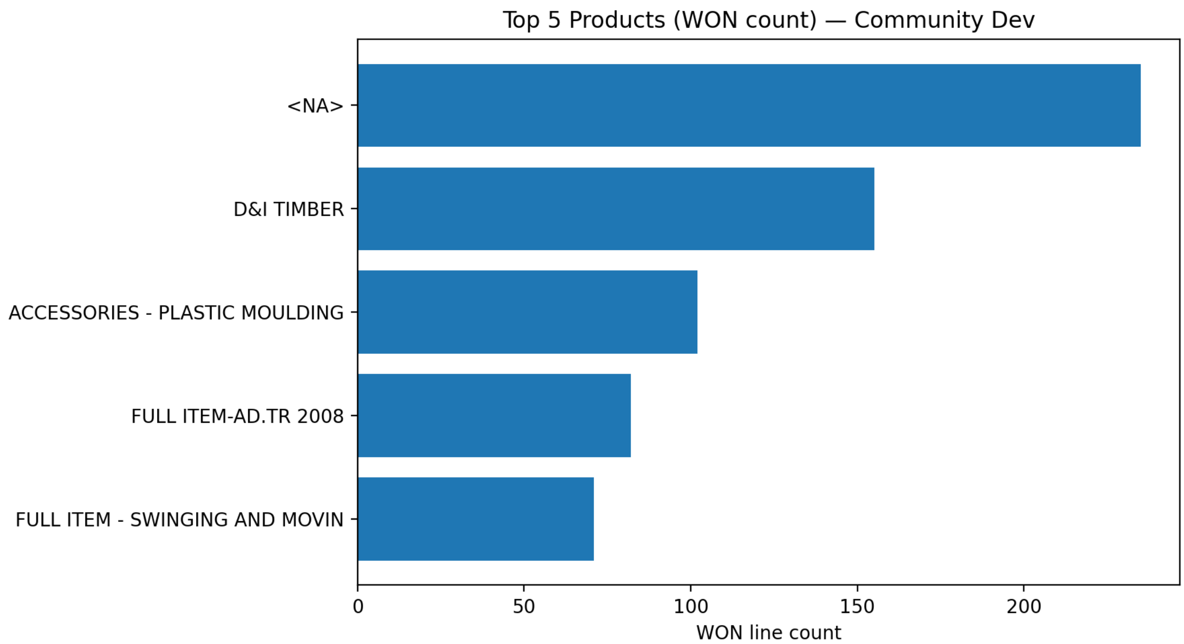


Figure A.3: Top Products for Community Dev.

# Appendix B

# Additional Tables

Table B.1: Model comparison on the held-out test set (threshold = 0.50).

| **Model** | **AUC** | **Acc** | **Lost** | | | **Won** | | |
|---|---|---|---|---|---|---|---|---|
| | | | **Prec** | **Rec** | **F1** | **Prec** | **Rec** | **F1** |
| ExtraTrees | 0.805 | 0.730 | 0.780 | 0.740 | 0.760 | 0.660 | 0.710 | 0.680 |
| Gradient Boosting | 0.793 | 0.720 | 0.820 | 0.670 | 0.740 | 0.630 | 0.790 | 0.700 |
| HistGradientBoosting | 0.801 | 0.730 | 0.820 | 0.690 | 0.750 | 0.640 | 0.790 | 0.710 |
| LinearSVC (calibrated) | 0.730 | 0.660 | 0.700 | 0.710 | 0.710 | 0.590 | 0.580 | 0.590 |